

(Final Spatial Dataset LOCKED)

What was built

Two files, joined by Ward_ID — deliberately NOT one giant merged GeoJSON:

File	Contains	Size	Rows
<code>data/processed/final_wards_static.geojson</code>	Ward geometry + population_density + ndvi_green_cover + vulnerability + response_capacity	244 KB	48 (one per ward)
<code>data/processed/final_ward_daily.csv</code>	UTCI/NCTL/DayStress/C TBI + vulnerability + priority_v2 + tier + drivers + recommended_actions	12.5 MB	26,208 (48 wards × 546 days)

Why two files, not one (architecture decision, not an oversight)

Attaching ward geometry to every date-row is **exactly the mistake that already happened once** on this project — Day 3's first version of `ward_thermal_daily.csv/.geojson` shipped at 96MB/132MB because geometry was duplicated 546 times per ward. Fixed then, and this final lock is built to never repeat it:

- **Vulnerability is static** (population/NDVI don't change day-to-day) → belongs with geometry, once per ward.
- **Thermal burden and priority are time-varying** → belongs in a plain CSV, joined back only when a specific date needs rendering.

This keeps the static file small (244 KB) and the daily file a manageable 12.5 MB — both comfortably under GitHub's 100MB limit, no Git LFS needed.

Fixed a real gap while assembling this

`action_engine_full_test.csv` (Khushi's Day 4 output) had dropped NCTL, UTCI_mean, DayStress on the way to producing action recommendations — fields the dashboard's detail card needs. `final_ward_daily.csv` re-merges these back in from `priority_rubric_v2_full_test.csv`, so nothing the dashboard needs is missing.

Verification (all passed, printed by the script)

- Static file: exactly 48 wards, all with real vulnerability + geometry ✓
- Daily file: exactly 48 wards × 546 dates = 26,208 rows, no missing/mismatched ✓
- Ward_IDs identical between both files ✓
- Sample join (latest date, static + daily merged): 48 rows, 0 missing CTBI ✓

For Vrinda — how to use this in the dashboard

python

```
wards_static =  
gpd.read_file("data/processed/final_wards_static.geojson")  
daily = pd.read_csv("data/processed/final_ward_daily.csv")  
  
# For any selected date:  
day_data = daily[daily["date"] == selected_date]  
gdf_day = wards_static.merge(day_data, on="Ward_ID", how="left")
```

This replaces `priority_rubric_v1_full_test.csv` (the stale source Shweta's Day 4 validation flagged — see `docs/Shweta_Day4_Validation_Report.md`).

`final_ward_daily.csv` already has the real vulnerability + priority_v2 + Action Engine recommendations built in, so this one change fixes the dashboard staleness issue in one step.

Lock notice

As of this commit, the spatial dataset is **locked**. Further changes to `final_wards_static.geojson` / `final_ward_daily.csv` should only happen for a

documented, serious bug — not routine edits — since Vrinda's dashboard integration now depends on this exact schema (column names, join key, row counts).

response_capacity — still honestly "Pending"

Healthcare-access (OSM hospital-distance) was never built. `response_capacity` is explicitly set to the string "Pending" in the static file — not fabricated, not silently dropped. Once it's built, it plugs into `priority_rubric_v2()`'s existing `response_capacity=` parameter with no schema changes needed.